then popular eugenics movement, Duncan said, 'Just imagine a child with my beauty and your brains!' France responded, 'Yes, but imagine a child with *my* beauty and *your* brains.') Well then, why not rig the lottery? Fertilise several eggs, and choose the one with the best combination. Once stem-cell research enables us to create an unlimited supply of human embryos on the cheap, you can select your optimal baby from among hundreds of candidates, all carrying your DNA, all perfectly natural, and none requiring any futuristic genetic engineering. Iterate this procedure for a few generations, and you could easily end up with superhumans (or a creepy dystopia).

But what if after fertilising even numerous eggs, you find that all of them contain some deadly mutations? Should you destroy all the embryos? Instead of doing that, why not replace the problematic genes? A breakthrough case involves mitochondrial DNA. Mitochondria are tiny organelles within human cells, which produce the energy used by the cell. They have their own set of genes, which is completely separate from the DNA in the cell's nucleus. Defective mitochondrial DNA leads to various debilitating or even deadly diseases. It is technically feasible with current *in vitro* technology to overcome mitochondrial genetic diseases by creating a 'three-parent baby'. The baby's nuclear DNA comes from two parents, while the mitochondrial DNA comes from a third person. In 2000 Sharon Saarinen from West Bloomfield, Michigan, gave birth to a healthy baby girl, Alana. Alana's nuclear DNA came from her mother, Sharon, and her father, Paul, but her mitochondrial DNA came from another woman. From a purely technical perspective, Alana has three biological parents. A year later, in 2001, the US government banned this treatment, due to safety and ethical concerns.[48]

However, on 3 February 2015 the British Parliament voted in favour of the so-called 'three-parent embryo' law, allowing this treatment – and related research – in the UK.[49] At present it is technically unfeasible, and illegal, to replace nuclear DNA, but if and when the technical difficulties are solved, the same logic that favoured the replacement of defective mitochondrial DNA would seem to warrant doing the same with nuclear DNA.